

40 The Evolution of Language Systems in the Human Brain

T W Deacon, University of California,
Berkeley, CA, USA

© 2009 Elsevier Inc. All rights reserved.

40.1 Introduction: Human Neural Language Adaptations	907
40.1.1 Language Uniqueness and Nonhuman Communication	907
40.1.2 Linguistic Context of Language Adaptation	907
40.1.3 Animal Exceptions and the Significance of Animal Language Experiments	908
40.1.4 Gestural Language: Neural Correlates and Evolutionary Scenarios	908
40.2 Human Neuroanatomical Features Associated with Language	909
40.2.1 Gross Neuroanatomical Homologies	909
40.2.2 Allometric Deviations Potentially Associated with Language Adaptation	909
40.2.3 Connectional Homologies and Dishomologies Relevant to Language	911
40.3 Comparative Functional Analyses	913
40.3.1 Functional Dissociation of Call and Speech Motor Control	913
40.3.2 Songbird Comparisons	914
40.3.3 Lateralization of Language Functions	914
40.3.4 The Mirror System	916
40.4 Genetic Correlates of Language Adaptation	917
40.4.1 Hopeful Monsters and Megamutation Scenarios	917
40.4.2 Genes Affecting Language Processing	917
40.5 Evolutionary Processes and Brain–Language Co-Evolution	918
40.5.1 Evolutionary Scenarios	918
40.5.2 Co-Evolutionary Scenarios	919
40.5.3 Degenerative Processes as Possible Contributors to Language Evolution	920
40.6 Conclusions	921

Glossary

<i>acheulean</i>	The stone tool technology associated with the hominid species <i>Homo erectus</i> .	
<i>akinetic mutism</i>	Immobility and nonresponsiveness due to dorsal frontal midline cortical damage involving the anterior cingulate cortex, that includes verbal nonresponsiveness.	<i>Broca's aphasia (area)</i>
<i>allometry</i>	The nonisometric scaling of anatomical structures with growth and comparative size.	
<i>arcuate fasciculus</i>	The fiber tract extending from the temporal and inferior parietal lobes to the inferior frontal lobes on the human brain passing beneath the supramarginal gyrus somatic and motor areas just dorsal to the Sylvian fissure.	
<i>Baldwin effect</i>	A theoretical evolutionary mechanism proposed independently by James Mark Baldwin, Conwy Lloyd Morgan, and Henry Osborne in 1896 arguing that physiological and behavioral plasticity could shield a lineage from elimination by natural selection long enough for new variations to accumulate (e.g., by chance mutation) that could supplement or replace the plastically acquired adaptation.	<i>co evolution</i>

shield a lineage from elimination by natural selection long enough for new variations to accumulate (e.g., by chance mutation) that could supplement or replace the plastically acquired adaptation.

The language disorder first described by Paul Broca in 1861 and elaborated in 1865 that was produced by damage involving the posterior part of the inferior third frontal convolution on the left side of the brain (commonly referred to as Broca's speech area). This aphasia syndrome is typically associated with nonfluent, telegraphic speech and often agrammatism (a difficulty constructing or assessing syntactic structures).

An evolutionary dynamic that is a consequence of two interacting selection processes, typically occurring between species whose survival is linked or between organizational levels of the biological hierarchy. In this context, it is

	used to describe the complex interactions that have likely characterized the evolution of the human brain and the evolution like processes of language transmission and change which may have imposed novel selection pressures on human brain evolution as an artificial niche.	<i>Homo rudolfensis</i>	One of the two earliest identified fossil members of our genus <i>Homo</i> dating to approximately 1.8 Mya in northeast Africa and associated with early stone tools. This species is characterized by a significantly larger brain case but minimally reduced dentition in comparison to earlier australo pitheciine hominids.
<i>diffusion tensor weighted MRI</i>	A structural MRI technique that uses oriented diffusion processes (constrained by fiber tract orientation) to visualize three dimensional organization of major axonal pathways.	<i>hopeful monster</i>	A hypothetical significantly deviant member of a species produced by a mutation that radically alters some phenotypic character which just happens to be better suited to the current environment and thereby manages to outreproduce and eventually replace the more typical phenotypes of the population. This theory was championed by Goldschmidt in the early 1900s and is often tacitly assumed by proponents of a saltational origin of language abilities.
<i>FOXP2</i>	A highly conserved transcription factor gene of the fork head family of genes that is associated with an inherited disorder of speech articulation and syntactic regularization.		The mode of reference that works by virtue of correlational relationships, as in pointings, symptoms, samples, and simple learned associations.
<i>generativity (generative grammar)</i>	The capacity for indefinite novelty of combinatorial uses of words provided by the grammatical and syntactic apparatus of a language. Generative grammars are theoretical rule governed grammars that by recursive application of these rules enable generativity of sentential forms.	<i>index (and indexicality)</i>	The mode of trait inheritance proposed by Jean Baptiste de Lamarck in the early 1800s in which physical and behavioral traits acquired by effort, exercise, or exposure to demanding conditions were presumed to be passed directly to offspring during reproduction.
<i>genetic drift</i>	The random mixing, accumulation of genetic variants, and elimination of alleles due to the relaxation of the effects of natural selection or the greater effect of probabilistic factors in small breeding populations.	<i>Lamarckian inheritance</i>	The stone tool industry associated with Neanderthals and early modern humans prior to the upper Paleolithic period beginning somewhere between about 60 and 75 kya.
<i>Homo erectus</i>	A long persisting fossil precursor species to modern humans (from approximately 1.6 Mya to 350 kya, depending on which specimens are included) with roughly modern stature and postcranial skeletal structure, and a brain size average in the range of 950 cm ³ (at the very low end of the modern range). <i>H. erectus</i> is found in Africa and also throughout Eurasia, extending into Europe, central Asia, China, and Indonesia.	<i>Mousterian</i>	The effect that a species has in altering its immediate environment so that the influences of natural selection are significantly affected by this modification, as in the way that beaver dam construction has played a significant role in providing selection favoring the evolution of aquatic adaptations.
<i>Homo habilis</i>	One of the two earliest identified fossil members of our genus <i>Homo</i> dating to approximately 1.8 Mya in northeast Africa and associated with early stone tools. This species is characterized by reduced dentition and a slightly larger brain case in comparison to earlier australo pitheciine hominids.	<i>niche construction</i>	The mode of reference that picks out objects of reference by virtue of a system of sign relationships (correlational relationships,
		<i>symbol (symbolic)</i>	

	as in pointings, symptoms, samples, and simple learned associations).
<i>universal grammar (UG)</i>	The hypothetical common core of grammatical rules that all human languages share. The theory was championed by the linguist Noam Chomsky and is argued by many linguists to be the innate endowment of all humans from which the specific grammars of existing languages are derived.
<i>upper Paleolithic</i>	An archeologically delineated period of human prehistory beginning roughly between 60 and 75 kya (most notably in Europe, but with more ancient precursors appearing in Africa) in which stone tool technologies begin to exhibit significant regional varieties and the first unambiguous representational forms (i.e., carvings and cave paintings) appear.
<i>Wernicke's aphasia (area)</i>	The language disorder first described by Wernicke (1874) that is produced by damage involving the posterior part of the superior temporal lobe on the left side of the brain (commonly referred to as Wernicke's speech area). This aphasia syndrome is typically associated with fluent speech that includes inappropriate, phonologically deviant, and/or semantically deviant word choice, and typically a deficit in comprehending sentence meaning and in naming objects.

40.1 Introduction: Human Neural Language Adaptations

40.1.1 Language Uniqueness and Nonhuman Communication

The comparative uniqueness of language is probably its most important and troubling feature. Besides being vastly more complex, language is substantially different in referential function, behavioral organization, and neural control than any other known animal communication system. Although a number of features are shared in common with the communication systems of some species, for example, vocal-auditory medium, social transmission, its most distinguishing characteristics – symbolic reference, grammar, open-ended generativity, and combinatorial patterning – are unprecedented. The lack of clear

behavioral homologies in other species renders the comparative method problematic. There are no other species with various grades of language to provide clues about the contexts that support language evolution (though there are species that exhibit the ability to acquire aspects of language; see below), or the range of brain systems that can be involved. There is only one exemplar, *Homo sapiens*, and if there were intermediate levels of these abilities in our ancestry they have all been eliminated. This noncomparability is made all the more enigmatic when we consider the apparent absence of neurological dishomologies with respect to major neuroanatomical structures that might be expected to correlate with such a significant cognitive-behavioral discontinuity. Generations of comparative neuroanatomists have failed to identify even one major novel brain structure in humans. This suggests that our special adaptations for language are the result of using previously evolved primate brain structures in new ways and in new combinations.

40.1.2 Linguistic Context of Language Adaptation

Because of this unusual status of language, it has long been regarded as one of the defining features of human distinctiveness. Historical efforts to explain its origins have consequently been confounded with efforts to define the essence of humanness. This tendency is well exemplified by linguistic debates about the origin and basis for language. Under the influence of persuasive arguments by the linguist Chomsky (1972) and the psychologist Lenneberg (1967), it became popular to argue that language depended on elaborate innate capabilities unique to humans. Chomsky has been particularly influential in articulating what this might entail. At the center of this theory is the claim that all humans have inherited a common innate universal grammar (UG). This innate faculty is presumed to make the acquisition of language possible even at a stage in the life when other forms of learning are undeveloped, and makes effortless the unconscious deployment of a vast set of syntactic rules. These rules are thought to underlie the real-time capacity to interpret or generate a nearly infinite number of grammatical sentences (generativity). Despite the fact that Chomsky, and other colleagues, locate this language capacity in the brain, he maintains the view that it cannot be explained as an adaptation, whereas other linguists (e.g., Jackendoff, 1994; Pinker, 1994) argue that it is an evolved adaptation (see Section 40.5). The formal tools developed by generative linguists over the past four decades have

provided unparalleled rigor for the analysis of morphology and sentence structure; however, despite their theoretical commitment to an inherited biological substrate for linguistic capacities, these methods have yielded relatively little in the way of verified neurological predictions, and it is also not clear that they could be substantiated or falsified by brain research. More than a generation has elapsed since this view achieved ascendancy; neither a discrete neural locus for grammatical processes or a neurological lesion that selectively disrupts core features of UG nor a genetic defect that produces systematically divergent forms of grammar have been identified (though neural and genetic impairments of certain features of morphological or syntactic processing have been identified; see below). One major reason for this may be the difficulty of translating highly abstract linguistic formalisms into concrete anatomical predictions. At least superficially, language appears to be generated according to symbolic principles that are very different from the phylogenetic and epigenetic principles that determine functional organization within brains. Nevertheless, linguistic lists of the necessary and sufficient capacities for language remain highly influential, and linguists have been the staunchest proponents of a radical discontinuity between humans and other animals with respect to language.

40.1.3 Animal Exceptions and the Significance of Animal Language Experiments

Efforts to identify analogues to human language features in nonhuman species' naturalistic communication have demonstrated only limited behavioral and functional overlap. The most influential examples include the vocal learning of parrots and songbirds, the socially transmitted songs of humpback whales, and the referential alarm calls of numerous species, but most notably vervet monkeys (see Section 40.3). Vocal learning is deemed significant because the vast majority of terrestrial mammals do not exhibit any significant capability to learn or mimic noninnate species-typical vocalizations. The examples of complex socially transmitted vocalizations in many bird lineages and in humpback whale pods thus exhibit a deviation from the norm that parallels a key characteristic of language. The referential function of alarm calls to pick out distinctive classes of predators has been demonstrated in primates, birds, and even rodents. Classic theories of animal calls had caricatured them as merely extrinsic symptoms of emotional states, so

demonstrations of specific extrinsic reference linked to specific innate calls also suggested parallels with the ubiquitous referential function of words and sentences. However, these superficial similarities are to be contrasted with many unprecedented language features.

Despite a failure to demonstrate any naturally occurring language-like systems outside of humans, partially successful efforts to train nonhuman species to perform certain limited language tasks have helped focus attention on the specific cognitive differences that separate them from humans. Studies of ape, dolphin, and parrot abilities to acquire language-like systems tailored for their different propensities and sensory-motor capacities have variously demonstrated the simple use of symbolic reference and a very basic understanding of syntactic operations, even if not anywhere near the level of interpretive and generative competence observed in a 3-year-old human child. Significantly, these three animal groups represent considerably different brain structures, since dolphin and especially bird brains (see below; and see *Do Birds and Reptiles Possess Homologues of Mammalian Visual, Somatosensory, and Motor Cortices?*, *The Evolution of Vocal Learning Systems in Birds and Humans*) are organized quite differently than are human brains. One possible implication is that at least rudimentary language-like capacities are not dependent on primate (or even mammalian) brain architecture, and so may be achievable in diverse ways.

40.1.4 Gestural Language: Neural Correlates and Evolutionary Scenarios

A counterpart to this diversity of potential neural substrates for language-like behaviors is demonstrated by the modality independence of language in humans. The manual languages that have developed in numerous deaf communities throughout the world show that fully complex languages can be acquired independent of the aural-vocal modality. However, despite this significant modality difference, there is also considerable overlap in neural representation with spoken language (Neville *et al.*, 1997). This may indicate that the critical neural adaptations supporting language are not, as is often suggested, merely specializations in motor or auditory processing, though these are also likely. Many scenarios for language origins have suggested that manual or gestural language preceded spoken language in evolution (see, e.g., Hewes, 1973; Corballis, 2002). Neurological support for this view comes from the absence of voluntary articulate control of laryngeal musculature in most terrestrial

mammals that have been studied, including all great apes (anatomy discussed below). Humans are the sole exception. It is therefore likely that the common ancestor of humans and chimpanzees lacked this control and that articulate vocal control is a derived trait that arose at some point in hominid evolution, possibly after the emergence of the genus *Homo*. Although most face-to-face speech is accompanied by gesture for emphasis and indexicality, language develops almost exclusively in the vocal channel, not manually, if speech and hearing are possible. Also, the locations of the major cortical systems critical for language processing are similar in both speakers and deaf signers. So support for a separate prior specialization of the brain for gestural language is weak. More likely, spoken and gestured symbolic communication were employed in linked fashion for a significant part of the evolution of language, with vocal capacities lagging behind but eventually becoming the more prominent modality.

40.2 Human Neuroanatomical Features Associated with Language

40.2.1 Gross Neuroanatomical Homologies

Generations of comparative neuroanatomists have explored the possibility that human brains contain species-unique large-scale structures (e.g., distinct nuclei, cortical areas, fiber tracts) that might correlate with our species-unique form of communication. However, since the famous nineteenth century debate between Thomas Huxley and Richard Owen in which Huxley disproved the existence of a uniquely human hippocampal structure, there has been widespread confirmation of the extensive homologies linking human and great ape brains and no verified claims of any phylogenetically unprecedented macroscopic structure in the human brain. Nevertheless, claims of the evolutionary functional divergence of brain structures are widespread in the literature. Two cortical regions have been consistently implicated in these claims: Broca's region in the inferior frontal lobes and the angular gyrus region at the temporal-parietal-occipital junction. The uniqueness of speech has led to hypotheses that Broca's region is uniquely developed in human brains, and the supramodal nature of semantic associations has led to hypotheses that a cross-modal association area is uniquely developed in the region of the angular gyrus. The hypothesis that Broca's region was uniquely developed in the hominid lineage led to investigations of fossil skull endocasts (see critical discussion below) and reports that the distinguishing sulci of this region could first be detected on an

endocast of a *Homo rudolfensis* specimen, KMNER 1470 (then identified as *Homo habilis*) (Falk, 1983; Tobias, 1987). Phyletic novelty of this structure has since been cast in doubt by the evidence of both cytoarchitectonic and connectional homologies of this region with corresponding regions in ape and even monkey brains (Deacon, 1992b; see Section 40.2.3). The hypothesis that the human angular gyrus region is an unprecedented cross-modal association area critical for language was first articulated by the neurologist Geschwind (1964). Early claims of poor cross-modal transfer of information in monkeys were subsequently disproven (Wegener, 1965; Blakeslee and Gunter, 1966), and subsequent studies have since identified polymodal function in homologous cortical regions as well as other inferior parietal and middle temporal areas (e.g., Ettlinger and Wilson, 1990). In response to this failure to find unprecedented brain structures relevant to human language facility, most attention has turned to quantitative, connectional, and peripheral dishomologies that may be relevant.

40.2.2 Allometric Deviations Potentially Associated with Language Adaptation

That the human brain has been subject to quantitative deviation from ape brain proportions is indisputable. Human brains are both absolutely and comparatively larger than expected for an anthropoid primate or even a great ape. For this reason, much attention has been focused on the plausible link between this deviation in brain proportions and the deviant features of human language. Both brain/body proportions and the internal scaling of brain structures with respect to each other and the gross brain size are highly correlated (Sacher, 1970; Gould, 1975; Finlay and Darlington, 1995), but most brain structures do not scale up or down isometrically with respect to total brain size across species. Allometric scaling patterns are also exhibited at every level of brain structure. For example, larger brains tend to have higher proportions of telencephalon to diencephalon, more neocortex to limbic cortex, more eulaminate cortex to specialized agranular and sensory koniocortex, more white matter to gray matter, more glia per neurons, and so on.

Generally, it is argued (on theoretical, not empirical grounds) that structural proportions that are predictable from allometric scaling (e.g., with respect to the trend exhibited by large interspecific sample as a background) indicate nondeviant function as well. Consequently, interest has mostly focused on quantitative findings of allometric deviation of human brain

structures with respect to apes or to anthropoid primates in general (e.g., Stephan, 1969; Stephan *et al.*, 1981). These investigations are complicated by disparities of results obtained using different statistical approaches, different methods of structural measurement, and disagreements about the significance of deviations that these analyses suggest. At present, there is no agreed upon theoretical basis (and limited empirical data) for predicting the functional correlates of either the allometric or deviant changes in relative proportions of brain regions. Comparative studies showing quantitative structural correlations with peripheral specialization offer the most useful comparisons. Examples of regional enlargements with respect to manipulative forelimbs (e.g., large forelimb tactile representation in primates and raccoons), elaborated or degenerate sensory organs (e.g., specialized tactile representation in the star-nose mole or elimination of visual cortical responses in the blind mole rat, respectively), or highly modified and hypertrophied organs (e.g., cerebellum in electric fish) offer support to the phrenological null hypothesis that increase in relative size equals functional augmentation as well. Perhaps those most relevant to the cognitive-behavioral specialization of language are the size correlations between song complexity in songbirds and the relative sizes of forebrain nuclei involved in singing (e.g., DeVoogd *et al.*, 1993). It is likely, however, that there are other possible correlates of allometric deviation that have yet to be explored (other alternative possibilities are explored in Deacon, 1990b).

Despite this uncertainty about the significance of allometric scaling and deviations in brain structure proportion, there has long been an interest in searching for possible correlations between allometric deviations of human brain structure and language. Different studies have, for example, provided analyses that suggest that human brains have divergent enlargement of cerebral and cerebellar cortices, prefrontal cortex, and certain thalamic nuclei, and divergent reductions of primary visual cortex, primary motor cortex, and olfactory bulbs. Similarly, quantitative studies have found hemispheric asymmetries in language-related areas of cortex, though such asymmetries have also been reported for nonhuman apes. Many of these findings must be considered preliminary, however, since most have been contradicted by studies using different methods that have come to different conclusions. There is also a considerable variation in the size of cytoarchitectonically identified language areas to contend with (e.g., Amunts *et al.*, 1999). One illustrative example of a quantitative dispute with implications for language concerns

the allometric predicatability or deviation of the human prefrontal cortex. Studies based on histological analyses of the cytoarchitectonic distinction between granular prefrontal and agranular premotor cortex have reported that human prefrontal cortex is allometrically larger than predicted with respect to other anthropoids (e.g., Deacon, 1997). In contrast, MRI-based studies using major sulci and fissures as morphological markers to discern frontal from parietal and temporal cortex suggest no deviation (Semendeferi *et al.*, 1997). Claims of disconfirmation of one or the other result are, however, clouded by the use of these different anatomical methods, different definitions of frontal and prefrontal cortex, comparison of nonhomologous structures, including different primate species as a comparison set, and employing different statistical tests for deviance (Deacon, 1997, 2004).

Despite these unresolved methodological issues, most studies have concluded that human brains deviate from allometric predictions in a number of internal relationships that might be relevant for language. Probably the most consistently reported finding is that human cerebral cortex is larger than allometrically expected with respect to the two major forebrain nuclear complexes that are most intimately related to it: the thalamus and the basal ganglia (Deacon, 1988, 1990a; Dunbar, 1993; Rilling and Insel, 1999). Additionally, there is evidence for allometric deviation of kinocortex and agranular cortex to eulaminate cortex (Deacon, 1990a), visual cortex (e.g., Holloway, 1979), temporal lobe morphology (Rilling and Seligman, 2002), and prefrontal cortex (Deacon, 1988, 1997; Rilling and Insel, 1999). It is hard to believe that significant deviations in these major forebrain relationships would not have an impact on language. For example, Deacon (1997) argues that this disproportion may have aided invasion of cortical efferents into brainstem vocalization nuclei (see below), as well as biasing developmental competition among cortical afferents affecting parcelation of functional cortical areas. Finally, difficulties of discerning comparable boundaries of cortical areas across species of widely differing sizes have made more fine-grained allometric studies of the scaling of individual cortical areas even more problematic.

Efforts to link allometric deviations of some of these structures with language adaptation have mostly focused on two findings, the expansion of prefrontal cortex (e.g., Aboitiz and Garcia, 1997; Deacon, 1997) and quantitative deviations and asymmetries of Broca's and Wernicke's language areas compared to their homologues in chimpanzees

(e.g., Gannon *et al.*, 1998). For example, prefrontal expansion may provide working memory support for symbol learning, visual cortex reduction may reflect parietal cortex expansion and an augmentation of cross-modal cognition, and asymmetries of language cortex may provide the substrate for hemispheric specialization for language.

40.2.3 Connectional Homologies and Dishomologies Relevant to Language

Neither the lack of novel human brain structures nor the uncertainties about the existence or relevance of allometric deviations precludes the possibility of evolutionary changes in neural circuitry. Unfortunately, methods used to accurately trace axonal connections between brain structures require lethal experiments and so are only available for study of nonhuman species, and indeed are not even applicable to apes because of their endangered status and the ethical issues involved. Thus, information concerning human neural connections is mostly lacking and must be extrapolated from nonhuman data. However, indirect evidence from human functional differences, clinical studies, and *in vivo* imaging can be compared to connectional data derived from nonhuman primates to support a handful of fairly robust connectional claims.

Probably the most robust behavioral distinction between the vocal abilities of humans and other primates (and in general with respect to all terrestrial mammals) is the ability for humans to produce a wide range of vocal sounds that can be freely organized into diverse combinations. In addition, we also have an unprecedented ability (for a terrestrial mammal) to mimic vocal sound combinations that we hear others produce. In contrast, the vast majority of terrestrial mammals, including primates, has relatively fixed vocal repertoires, for which sound mimicry learning plays almost no role. Associated with our lack of constraint on productive sound combination, we experience a relative freedom from specific correlations between vocalizations, emotional states, and stereotypic referential contexts, unlike what is characteristic of the other primates (Deacon, 1997). This is a critical requirement for language, as it allows socially transmitted patterns of sound production (e.g., words) to be learned in association with any given reference. Though humans still exhibit a small repertoire of innate stereotypic species-specific vocal 'calls' such as laughter and sobbing, which do have fixed structure and are associated with highly constrained emotional contexts, this call repertoire is both small compared

to that in chimpanzees and atypical in form and context (Deacon, 1997; Provine, 2000).

These differences are probably in part attributable to a change in the central innervation of the laryngeal control nucleus of the brainstem, the nucleus ambiguus. Tracer studies in nonhuman primates have demonstrated that this nucleus is almost entirely innervated by subcortical structures from midbrain and adjacent brainstem regions (Jürgens *et al.*, 1982). This is an expected pattern given that the nucleus ambiguus is a visceral motor nucleus that is segregated from significant influence from volitional systems in order to provide reliable automatic responses for a system that is associated with life-and-death consequences. Though electrical stimulation of ventral motor cortex regions in the macaque monkey brain can result in vocal muscle movement, there is little evidence that this is mediated by a direct projection. In addition, bilateral ventral frontal motor cortex damage in monkeys does not appear to block their ability to vocalize. In contrast, unilateral damage to left inferior motor cortex in humans (even just in the left hemisphere) can produce significantly impaired vocal ability, and even mutism. This clinical evidence is supported by experimental studies that also suggest that the human nucleus ambiguus is directly innervated from motor cortex (Jürgens *et al.*, 1982). Taken together, this makes it likely that this connection constitutes a uniquely human feature, by virtue of which precise control of pitch and vocal timing is achieved in speech and song. This would also explain the remarkable coordination of vocalization with the other cortically controlled tongue, jaw, and facial muscles that are necessary for articulate speech. In this regard, humans have dual control of vocalization, as is exhibited in the tendency for speech to be interrupted with impulses to laugh or sob in response to intense emotional states (Provine 2000; see Figure 1).

The left inferior frontal cortical region that likely includes Broca's speech area has been subject to conflicting claims concerning (1) which components of this region are responsible for the deficits associated with Broca's aphasia, (2) how they are functionally connected with other cortical and subcortical regions also associated with language processing, (3) whether the cortical area itself or its connections with other areas are more important for its language role, and (4) whether or not its connectivity with other structures is typical of other primate brains. The clinical literature is even still split on what Brodmann's areas (Brodmann, 1909) are the substrates for Broca's area language

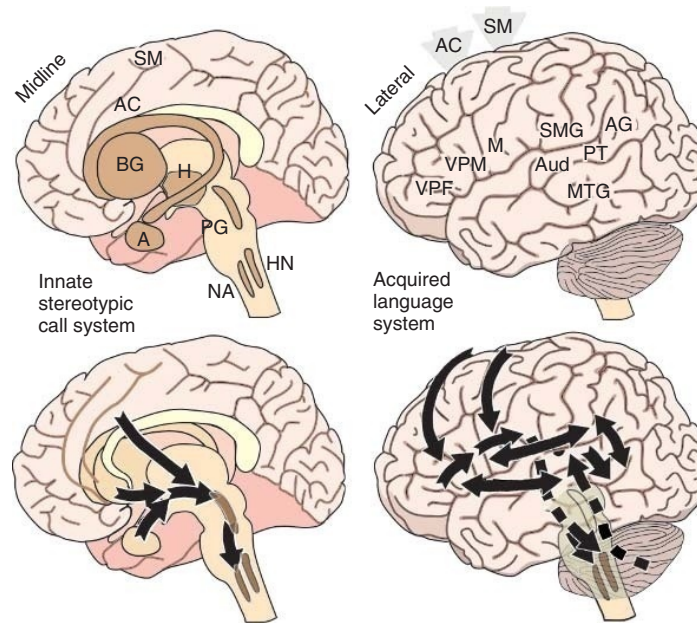


Figure 1 Highly simplified schematic comparison of brain structures (above) and major connections (below) comprising the mammalian innate call production system and the human spoken language system. Many connections and relevant structures are not shown, including notably the involvement of a cortical basal ganglia thalamic cortical loop and the cerebellum in language production. The forebrain structures of the innate call system are almost exclusively associated with arousal control, whereas those supporting language are almost exclusively sensorimotor and 'association' cortical structures. So these two vocal communication systems of the brain are largely nonoverlapping in terms of structures and connections, with the exception of final brainstem output systems controlling oral, vocal, and respiratory muscles, and the anterior cingulate cortex. Projections from motor cortex extend directly to brainstem vocal motor nuclei, whereas forebrain output controlling innate calls is mediated by the periaqueductal gray area of the midbrain. In humans, both systems operate in parallel, and may compete for the control of vocal output. The differential involvement of numerous interconnected forebrain systems in language as compared to the few involved in innate vocalizations is superficially similar to the neural differences between birds that learn complex variable songs and those with innate stereotypic songs (see Figure 2). A, amygdala; AC, anterior cingulate cortex; AG, angular gyrus; Aud, auditory area; BG, basal ganglia; H, hypothalamus; HN, hypoglossal nucleus; M, motor cortex; MTG, middle temporal gyrus; NA, nucleus ambiguus; PG, periaqueductal gray; PT, planum temporale; SM, supplementary motor area; SMG, supramarginal gyrus; VPM, ventral premotor; VPF, ventral prefrontal.

functions (Dronkers *et al.*, 1992), whether multiple frontal cortical areas subsume component language functions (Paulesu *et al.*, 1997; Deacon, 2004), and whether these cortical areas are the primary locus, rather than the underlying white matter and striatal structures (D'Esposito and Alexander, 1995; Lieberman, 2002). Nevertheless, claims that this area is in some way uniquely organized in humans are reinforced by the common incidence of agrammatism in Broca's aphasics and by the belief that it is grammar that sets humans apart from the other species.

One of the long-standing assumptions about Broca's area is that it is a convergence zone where auditory input contributes to the formulation of speech. Could this connection pattern be a unique feature of human brains supporting speech? Combining connection data from primates with new *in vivo* functional image data on language processing, it is possible to settle some of

these long-standing questions. Tracer studies of connections of the macaque inferior frontal cortex demonstrate linkages with other cortical and subcortical sites that include both parietal and temporal cortical areas. But these tracer results delineate at least three quite different connection patterns associated with different subregions of the monkey ventral frontal cortex (Deacon, 1992b). Motor and premotor cortical areas are primarily interconnected with inferior parietal and superior insular regions of cortex, and premotor cortex is connected with dorsal midline supplementary motor cortex. In comparison, the rostrally adjacent ventral prefrontal area is primarily interconnected with superior temporal and middle temporal gyrus areas, as well as with dorsal prefrontal areas and anterior cingulate cortex, but lacks connections with motor areas. So primate connection data do not support a simple convergence of auditory and tactile motor

functions in their anatomical homologue to Broca's region, and instead exhibit a tier-like organization with caudal-rostral segregation of parietal from temporal input zones. Physiological confirmation of an auditory projection zone in this macaque ventral prefrontal region has been provided by single cell recording (Romanski *et al.*, 1999). But is this segregation of auditory and motor functions in the primate homologue to Broca's area evidence that monkeys and humans differ in this respect? The corresponding connection patterns in the human brain have recently been traced using diffusion tensor weighted MRI techniques, which enable the visualization of fiber tracks (Catani *et al.*, 2005). This study mapped the course of the components of the fiber bundle known as the arcuate fasciculus, which in humans carries fibers presumed to interconnect Wernicke's area with Broca's area. The findings are consistent with the monkey brain connection pattern, not with a simple convergence zone logic. Inferior parietal projections terminate in ventral motor and premotor areas and superior and middle temporal gyrus projections terminate more rostrally in ventral prefrontal areas. Inferior parietal areas and superior temporal areas are also interconnected, but not superior temporal areas and ventral motor or premotor areas. As in the macaque brain, auditory information is relayed to the frontal areas by way of a prefrontal cortical area in front of and separate from the premotor-motor areas involved in speech production.

This evidence for fractionation of the contributions to language processing in ventral frontal cortex is also consistent with the accumulation of *in vivo* imaging data that show slightly different localizations in this region for heightened activity during language tasks that differentially involve auditory and motor processing. For example, word association and linguistically mediated mnemonic tasks preferentially activate the ventral prefrontal component, while tasks involving motor analysis preferentially activate premotor and motor areas located more caudally. The implications are first that Broca's area is not a single functional unit, but comprises two or more adjacent regions, second that only the prefrontal component utilizes temporal auditory input, and third that the language specialization of this region did not depend on any major restructuring of connectivity. In addition, if as in macaques this same auditory recipient ventral prefrontal area is linked to the anterior cingulate cortex, it would also represent a bridge between a language-specialized area and the one cortical

area known to be involved in primate call production.

40.3 Comparative Functional Analyses

40.3.1 Functional Dissociation of Call and Speech Motor Control

The discovery of predator-specific alarm calls in vervet monkeys (Seyfarth *et al.*, 1980) suggested that the functional dichotomy between language and primate call systems might not be so great as once believed. The existence of distinct calls given to leopards, eagles, and snakes suggested that the origins of language might be envisioned as a gradual elaboration of a larger and larger specific repertoire eventually requiring more complex production and combination mechanisms. However, evolutionary continuity is difficult to support when the difference in neural substrates between calls and language is considered. Electrical stimulation and lesion experiments established that primate calls could be elicited by stimulation of midbrain and limbic forebrain structures, including basal forebrain, ventral striatum, amygdala, hypothalamic, and anterior cingulate cortex, but not by cerebral cortical areas (e.g., Jürgens, 1979). Correspondingly, damage to monkey's cerebral cortical areas homologous to those involved in language processing in humans do not interfere with call production. Conversely, damage to limbic and telencephalic structures homologous to structures supporting primate calls do not produce language deficits in humans. There are some exceptions that prove this rule. One is the anterior cingulate cortex, which if bilaterally damaged in humans, may result in akinetic mutism (immobility that includes vocalization). But this is arguably an impairment of the arousal to speak and move rather than a disturbance of language processing. There have also been reports that stimulation of the amygdala in human subjects can sometimes produce spontaneous curses as well as emotional cries. And in patients with global aphasia, cursing is sometimes spared or even facilitated. Expletives are, however, an interesting intermediate; an acquired vocalization that has become relatively automatic and stereotypically associated with specific intense emotional experience. Taken together, these data demonstrate that the neural substrates for language functions and innate calls derive from almost completely dissociated brain systems. Along with evidence that speech depends on direct cortical projection to oral-vocal motor nuclei, independent of the older

limbic-midbrain-brainstem pathway, and the fact that speech and human innate calls exist side by side in our vocal repertoires, we can confidently conclude that the language system is not an elaboration of the call production system. The only significant overlap of these two systems is a final common output pathway.

40.3.2 Songbird Comparisons

Despite the fact that telencephalic organization in birds and mammals is radically different, there are useful analogies that can be drawn from comparison to birdsong control and its differences in different species (Jarvis, 2004). Research into the organization of song acquisition and control in different bird species demonstrates a consistent pattern that distinguishes song learners able to produce complex songs from nonlearners with simple songs. Comparisons between songbirds, parrots, and hummingbirds also demonstrate that complex singing abilities have evolved independently at least three times in the course of bird evolution and that in each of these lineages motor control of song output is mediated by slightly different forebrain systems. Species with highly stereotypical innate songs utilize only one or two forebrain motor nuclei for song production. In both songbirds and birds with stereotypic songs, a primary motor output nucleus in the caudal telencephalon (RA in songbirds) projects to the common vocal output pathway in central midbrain, and from there to brainstem motor nuclei. In addition, species that learn significant aspects of their songs and produce complex variable songs may require the coordinated contributions from as many as a dozen forebrain structures. In this regard, the difference between birds that sing complex sounds and those that sing stereotypic songs is crudely analogous to the human/nonhuman primate difference. So understanding the differences between the alternative complex song control strategies in different bird lineages and the difference between complex singers and stereotypic singers may provide useful comparisons (see Figure 2).

Two major classes of forebrain systems are integrated with the forebrain motor output nuclei to enable song learning and song complexity: auditory and striatal motor systems. These are necessary for learning from auditory experience. In addition, a higher-order premotor nucleus (high vocal center (HVC) in songbirds) is also critical for song complexity and flexibility. The differences between different lineages where vocal learning evolved are also interesting as a comparison. Some of the most sophisticated learners, such as parrots, have a

different motor output pathway from the forebrain than do oscine songbirds. Although it is not clear from current research whether these differences are more than variations on a theme, differences in the midbrain/brainstem output targets of these nuclei are suggestive. For example, in songbirds, a midbrain region (probably homologous to the mammalian periaqueductal gray area which mediates call production) mediates between forebrain and brainstem motor control nuclei, but in parrots and possibly hummingbirds there is also a direct projection from forebrain motor nuclei (not homologous to RA) to brainstem motor systems. It is tempting to speculate that this latter pattern is analogous to the direct forebrain (cortical) projections to the vocal motor centers that distinguishes humans. Whether these differences account for the remarkable vocal mimicry of parrots and their kin is unknown.

So, although a good deal is still to be learned about the evolution of vocal learning and vocal skill in birds, exploring this more experimentally accessible parallel to the human case may provide important clues about how vocal flexibility evolves. An intriguing example is discussed below.

40.3.3 Lateralization of Language Functions

One of the more enigmatic features of language representation in the brain is the fact that the two hemispheres play very different and unequal roles in controlling speech and comprehension. This is not because there are different cortical areas on the two sides. Ever since the French surgeon Paul Broca first catalogued cases of speech impairment associated with localized brain lesions (Broca, 1865), it has been known that the left-hemisphere damage is far more debilitating for language functions than the right-hemisphere damage. The brain regions in the inferior frontal gyrus and superior temporal gyrus that are the loci most likely damaged in Broca's and Wernicke's aphasia, respectively, are identified for the left hemisphere, but their right-hemisphere counterparts can often suffer damage with no obvious speech impairment. This left 'dominance' for language, as it is often described, is not universal, with just a few percent of people exhibiting complete right-sided language bias. The exceptions also correlate strongly with left handedness, suggesting a link between the asymmetrical biases. Beginning in the 1960s, a series of surgical interventions to limit the spread of epilepsy susceptibility from one hemisphere to the other cut the corpus callosum, and other forebrain commissures, severing the two hemispheres so that they could not

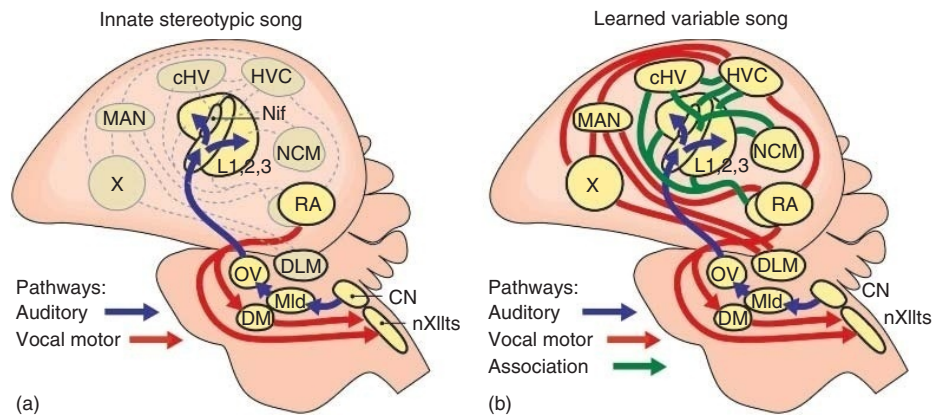


Figure 2 Comparison of major structures and connections in brains of (a) songbirds that do not learn their song and (b) those that do. Dashed connections and structures not connected by heavy lines are minimally involved in song (many structures and connections are not shown for simplicity). Innate stereotypic birdsong is relatively insensitive to damage to most forebrain structures (a) excepting a primary motor nucleus (RA) which projects to the central midbrain and to vocal motor nuclei, such as the hypoglossal nucleus (nXIIIts). In contrast, striatal structures (e.g., areas X and MAN), auditory centers (e.g., L1, L2, and L3), and numerous pathways linking these with premotor (e.g., HVC) and motor (RA) nuclei are all involved in various aspects of learning a song 'dialect' from adult singers in early life. In these birds damage to any of these structures or their interconnections can disturb learning, complexity, or flexibility of song. Other bird orders exhibiting learned vocalizations, such as parrots and hummingbirds, are distinguished by differences in final motor output pathways (not shown). The distribution of song control to a diverse system of forebrain structures in song learners is loosely analogous to the shift in control from limbic structures to the diverse system of interconnected sensory, motor, and association cortical areas and striatal nuclei that evolved to control language (see Figure 1). In both systems the involvement of a diverse constellation of interconnected forebrain structures appears to be correlated with complex, flexible, context-sensitive, socially transmitted, learned vocal skills. For a more detailed account of the comparative neurology of bird vocalizations and human language see Jarvis (2004). cHV, caudal region of the hyperstriatum ventrale; CN, cochlear nucleus; DM, dorsal medial nucleus of the midbrain; DLM, medial nucleus of the dorsolateral thalamus; MAN, magnocellular nucleus of the nidopallium; HVC, high vocal center; L1-3, primary auditory fields; Mld, mesencephalic lateral dorsal nucleus; NCM, caudal medial nidopallium; Nif, interfascicular nucleus of the nidopallium; nXIIIts, hypoglossal nucleus (nucleus XIIth cranial nerve, tracheal syringeal division); OV, nucleus ovoidalis; RA, robust nucleus of the arcopallium; X, area X of the striatum.

exchange signals. The results were startling. If information was carefully provided to only one of the isolated hemispheres, patients could use language to describe the stimulus only if presented to the left hemisphere (input from the right side). This suggested that the right hemisphere was essentially mute.

There are a few clues to how and why this functional asymmetry evolved to be so robustly associated with language functions. The first clue comes from understanding what language-related functions, if any, are contributed by the contralateral counterparts to Broca's and Wernicke's areas in the right hemisphere. Two lines of evidence suggest that, contrary to earlier views, the right hemisphere does indeed contribute to language processes. Both kinds of evidence come from cases of right-hemisphere brain damage. First, there is a higher incidence of aprosodia with right-hemisphere damage (Ross, 1981). Aprosodia is an impairment of the ability to produce or accurately comprehend the changes in

tonality and rhythmicity that is used to convey emotional tone, emphasis, or differential focus in speech. This suggests that the right hemisphere is involved in regulating the nonreferential social-emotional context of spoken conversation in parallel with the left-hemisphere production and comprehension of the syntactical and semantic content of speech. Second, right-hemisphere damage appears to impair the ability to keep track of the larger semantic and pragmatic frames of speech. Right-hemisphere-damaged patients have difficulty understanding what makes one joke funny and another lame, and also have difficulty following the theme of a story, often failing to recognize the insertion of incongruous elements (Gardner *et al.*, 1983). So although the right hemisphere appears to be minimally, if at all, involved in immediate semantic and syntactic processing of words and sentences, it appears to be carrying out important supportive background tasks in parallel.

This parallelism may help to explain another curious feature of language lateralization: its

development in childhood. Studies of very young children who have had their left hemispheres surgically removed show a remarkable sparing of language abilities, with minimal obvious impairment in adulthood (see review in Kolb, 1995; and recently Boatman *et al.*, 1999). So, although left lateralization of the semantic, syntactic, and phonological processing of language is highly predictable and probably reflects an innate bias, it is only a bias and not a fixed and inflexible adaptation.

Why should there be laterally asymmetric distributions of language functions in an otherwise bilaterally symmetric brain? Although there is no certain answer to this question, considering the functional characteristics that are lateralized, the fact of their progressive differentiation during maturation, and other correlates of lateralization (e.g., handedness) some plausible hypotheses can be formulated. First, there are consistent structural asymmetries in human brains (discussed already). The relative sizes of cortical areas and morphological structures associated with Broca's and Wernicke's language areas indicate that the right side counterparts are on average smaller. This could be a source of developmental bias or also a consequence of developmental differentiation, but other left-right asymmetries in neonatal brains support the possibility that anatomical bias contributes. Second, the language functions that appear to segregate to opposite hemispheres seem to divide according to both rate of processing (rapid processing on the left) and extent of conscious online monitoring required (also left). This can probably be understood in terms of segregating functions that need to run in parallel but would likely interfere with one another because of their very different processing parameters (Deacon, 1997). Third, the correlation with asymmetric manual skill suggests a possible linkage, perhaps with tool use (Kimura, 1993). Finally, the unilateral representation (also to the left) of vocal skill learning is also characteristic of songbirds (Nottebohm and Nottebohm, 1976). Since vocalization involves muscle systems that are aligned along the midline of the body and are bilaterally controlled, it may be necessary to strongly bias control to one hemisphere in order to avoid functional conflict. In summary, lateralization may not be a requirement for the evolution of language, but it is likely a bias built in to aid functional segregation of processes that are best run in parallel systems and thus avoiding mutual interference.

40.3.4 The Mirror System

Although the class of cells called 'mirror neurons' are discussed elsewhere in this volume, including their possible roles in language processing, this class of neural responses has also been implicated in many language origins theories (Rizzolatti and Arbib, 1998). Recording from single neurons in a ventral premotor subregion (designated F5) of the macaque monkey brain, Rizzolatti and colleagues identified a subset of neurons that preferentially spiked when the subject observed himself picking up an object and also when observing an experimenter picking up the same object in the same way. This responsiveness to the general form of the action, irrespective of the role of agency and perspective, suggested the name 'mirror neuron'. The relevance to the evolution of the language capacity is twofold: first it suggests the possibility that this kind of neural responsiveness could play a role in the ability to mimic others, and second the location of these cells in the macaque brain is in a region generally considered adjacent to the monkey brain region deemed homologous to the premotor division of Broca's area (see Section 40.2.1), and possibly overlapping. *In vivo* imaging data have further suggested that there is a similarly responsive premotor region in the human brain. Although it can be debated whether this response characteristic is specifically found in the same premotor region in human brains as the one that plays a critical role in language, these coincidences make it reasonable to entertain the hypothesis that this might help support the vocal mimicry necessary for word learning in language. If so, what might be the implications for language evolution? First, it must be noted that the presence of mirror neurons in macaque brains is sufficient to exclude these cells or their connections from being the difference in human brains that makes language possible; however, it could be argued that their presence predisposed this premotor zone for a later role in language processing. Second, speech mimicry demands that a parallel class of auditory-vocal mirror neurons be identified (visual-manual mirror neurons might be more important for mimicry in gestural language), though some of these neurons exhibit responses to the sound of an object being manipulated as well. In the monkey brain, mirror neurons receive input from neurons in inferior parietal cortex that are also responsive to visuo-manual stimuli, but if there are corresponding auditory-vocal mirror neurons we might rather expect them to receive input from superior or

middle temporal sources. Until such a parallel class of neurons is identified it is probably premature to assume that mirror neurons are critical to language functions, but looking for them is thus a relevant enterprise. One plausible scenario – assuming that mirror neurons are indeed critical for mimicry – is that they played a role in an early, more gestural phase of language evolution, and possibly paved the way for the evolution of these hypothetical auditory counterparts.

40.4 Genetic Correlates of Language Adaptation

40.4.1 Hopeful Monsters and Megamutation Scenarios

Probably the most popularly accepted scenario for language evolution is what has sometimes derisively and sometimes seriously been referred to as the ‘big bang’ scenario. On the analogy to the birth of the universe, this scenario suggests that language was made possible as a result of one or just a few major mutation events that resulted in the significant reorganization of brain functions. This resonates well with assumptions about an innate UG (see Section 40.1.2) or a ‘language acquisition device’ constituting the difference between human and nonhuman brains. It also resonates with paleoarcheological theories for explaining the sudden burst of cultural artifacts (such as diverse tools types, representational cave paintings, and carvings) that arose within the last 50 000 years. Since *H. sapiens* has been around for greater than 100 000 years, and hominids with comparably large brains and complex stone tools have been around for roughly half a million years, this transition appears quite recently in human evolution. But the idea that what distinguishes speaking humans from other species and from our recent ancestors can be explained by a couple of very lucky genetic accidents seems both counterintuitive in terms of what is known about the genetics of the developing brain and what is known about the complexity of language control. But more generally, it also leaves almost the entire explanation of this adaptation to an incredibly lucky accident. This kind of evolutionary scenario is often described as a hopeful monster story, because it imagines that a mutation producing a major phenotypic distortion becomes so enormously successful that it replaces all alternatives. Though one cannot argue that it is impossible, it is a claim about evolution that is little better than

invoking a miracle. Nevertheless, there is at least one serious proposal for just such a critical genetic change.

40.4.2 Genes Affecting Language Processing

In the mid-1980s, when excitement about the plausibility of innate UG was at its peak, a surprisingly specific inherited language disorder was described. Called specific language impairment (SLI) by researchers, it was expressed in a family (identified as the KE family) in which many members exhibited specific difficulty with regularized aspects of English syntax (Gopnik, 1990; Gopnik and Crago, 1991). Most notably, this was manifested in a problem learning to use the regular past tense ending ‘-ed’. Subsequent study of this deficit showed it to also be accompanied by significant oral motor apraxia (pronunciation and fluency problems) and neurological reduction of motor areas and basal ganglia (Vargha-Khadem *et al.*, 1998). Chromosomal damage to a common locus was subsequently correlated with expression of this trait, and in 2002 a transcription factor gene, FOXP2, was identified as the critical damaged gene in this disorder (Enard *et al.*, 2002), expressed in structures affected in the KE family (Lai *et al.*, 2003). It is the first single gene to be correlated with a known neurological disturbance of language function. It is not a ‘new’ gene unique to the human lineage, since it is present and plays a critical role in development of the brain in all mammals (a homologue is also found in birds and fruit flies), and it is a highly conserved gene in terms of sequence variations across species, and the KE family variant is damaged at a site conserved in all known species (and so likely critical).

Important with respect to its plausible role in the evolution of language are two point mutations in the human variant that distinguish it from the chimpanzee version (and basically from all other mammals in which it is highly conserved). Linkage information even suggests that these human deviations are relatively recent – possibly within the last 100 000 years – and are likely universal or nearly so in living humans. This does not prove, however, that these human-specific differences contribute to a crucial change in function (though the evidence is highly suggestive), and the alterations do not correspond to the damaged locus in the KE family. At the present time, we cannot even say for certain that having the chimpanzee gene would result in diminished language function, or whether a chimp with a human version of the gene would have improved oral motor capacity. But damage to a regulatory gene that is critical for early brain development (as it is in all

mammals) will almost certainly result in significant disruption of function, since it likely controls the expression of many other genes. So although the gene did not evolve for language, the neural features it controls during development have clearly been recruited by language, and it seems likely that the mutations that occurred in it in human evolution played some role in the evolution of speech.

Assuming that the point mutations of FOXP2 that are unique to the human lineage do play a role in our language adaptation, we next need to ask what kind of effect. And this requires considering its contribution to development of specific brain structures. Comparative and clinical data suggest that it plays a role in the development of the basal telencephalon, which will become ventral basal ganglia and basal forebrain in the adult. Though these basal ganglia structures are not classically identified as language structures *per se*, there are many reasons to think that basal ganglia structures could be important contributors to language learning and use (Lieberman, 2002), particularly of those processes that become relatively automatic. This is consistent with the critical role played by basal ganglia in skill learning, the automatization of many routine behavioral functions, and the establishment of procedural memories. Since there is an extensive interdependence between the anterior cortical areas and basal ganglia, via a recurrent circuit through the pallidum and thalamus, it should be no surprise that functions associated with frontal language cortex might also be affected by basal ganglia disturbances, especially motor functions. As a comparison, disruption of fluency, pronunciation, and syntactic processing have all been shown in Parkinson's disease patients, who also have reduced basal ganglia function (Lieberman, 2002). So if recent human-specific point mutations in the gene FOXP2 do reflect an adaptation for language processing, it is most likely with respect to aiding speech automatization. This role in learned vocalization is also supported by two additional comparative findings. First, the bird homologue to FOXP2 is found to be expressed in striatal nuclei associated with song learning (particularly in area X), and is more extensively expressed in species that learn their songs (Haesler *et al.*, 2004). Second, damaging one copy of FOXP2 in mice produces an impairment of their ultrasonic vocalization, and damaging both causes severe motor impairments, elimination of ultrasonic vocalizations, and premature death (Shua *et al.*, 2005). So although it is not specifically a gene for language, nor did it evolve only in humans, it has

clearly been critical for neural systems underlying vocal motor functions in terrestrial vertebrates for a very long time, and it may have been tweaked in recent human evolution.

40.5 Evolutionary Processes and Brain–Language Co-Evolution

40.5.1 Evolutionary Scenarios

Estimates of the age of language date from as little as 50 000 years to more than 2 million years. Some of this difference reflects different definitions of language, some reflects different notions about the tempo of evolution (i.e., whether the change was sudden or gradual), and some takes different views about the number of mutational changes that were necessary. In general, those who argue that the language faculty is a highly specialized modular capacity tend to favor a recent date of origin and a saltational transition, whereas those who favor a more generalized conception of the language faculty supported by a constellation of adaptations tend to favor more ancient dates.

Exactly how the processes of natural and sexual selection might have contributed to the evolution of the human language adaptation is also contentious. Darwin (1871) argued that language might have evolved from something like courtship song under the influence of sexual selection. Modern theories that appeal to sexual selection have also focused on the use of language for social manipulation. The most common scenarios, however, focus on the role of language as a tool for social coordination and maintenance of social groups.

Two extreme language selection scenarios are commonly opposed in the literature to predict what changes in brain structure might be relevant: scenarios assuming that language is a consequence (or late-stage tweak) of a more prolonged trend toward increasing general intelligence (exemplified by a 2-million-year expansion of brain size) and scenarios assuming that language is the consequence of domain-specific neural modifications and is independent of general intelligence. These are not mutually exclusive options, but they do make different predictions with respect to neural structural and functional consequences, as well as evolutionary timing. These functional implications can be used retroductively to probe the plausibility of each. If language has an ancient origin, it would follow that it is likely supported by a significant and extended natural selection history, including the contributions of many genetic changes affecting

the brain. If, on the other hand, language is of recent origin and largely without precedent, it would follow that little time has elapsed for significant effects of natural selection to accumulate. As a result, ancient origin hypotheses predict that language functions will be more thoroughly integrated into other cognitive functions, will likely have distributed representation in the brain, and should be highly plastic with respect to both minor brain disorders and genetic variation. Recent origin hypotheses, on the other hand, are more consistent with language processes being highly modular and domain specific, localized to one or a very few neural systems, fragile with respect to brain damage and genetic variation, and possibly radically altered in grammatical organization by genetic abnormalities. With the exception of claims for domain specificity, which are controversial, the neuropsychological evidence argues against a recent rapid transition to language capacity. But archeological evidence is also brought to bear on this question. The paleoarcheological record is surprisingly stable from about 1.6 Mya to roughly 350 kya, with the transition from Acheulean to Mousterian tool culture, but does not begin to show signs of regional tool styles, decorative artifacts, and representational forms (e.g., carvings and cave paintings) until roughly 60 kya, with the dawn of what is called the Upper Paleolithic culture. This recent transition to technological diversity and representational artifacts has been attributed to a major change in cognitive abilities, which many archeologists speculate reflects the appearance of language. Fossil crania, however, provide no hint of a major neuroanatomical reorganization, and the genetic diversity of modern human populations indicates that there are some modern human lineages who have been reproductively separated from one another for at least twice this period and yet all have roughly equivalent language abilities. These considerations weigh in favor of a protracted evolution of language abilities and for the convergence of many diverse neural adaptations to support language (Johansson, 2005).

An adaptive convergence logic also helps to resolve some of the mysteries concerning the absence of direct neuroanatomical or functional homologies between language and nonhuman communication adaptations. The novelty of language can be understood in terms of the combined effects of systems which individually may have served quite different functions in ancestral species but which collectively interact in novel ways to produce emergent consequences. If the human language

adaptation reflects the combined contribution of many diverse systems whose parallel evolutionary paths have come together to provide an unprecedented functional synergy, we should not expect to find highly divergent local changes in brain structure, but rather global reorganization in which most structures participate in some respect or other. But considering language functions to be emergent adaptations, in this sense, poses new questions about the evolutionary process. Specifically, we must explain how such functional synergies among diverse systems can be explored and recruited by the process of natural selection. In general, this reflects a common challenge posed to evolutionary theory since the time of Darwin, and can be generally answered the way he explained the probable evolution of the eye. He argued that even quite minimal non-image-forming light-sensitive proto-eyes would, none the less, provide an adaptive advantage over the absence of any light sensitivity, and that any minor modifications to adjacent structures that improved on this in any way would likewise be advantageous and selectively retained. As more comparative anatomical and genetic information has come to light concerning the evolution of eyes, in the century and a half since Darwin's time, his speculation has found ample justification. However, language differs from this sort of complex adaptation in one important respect: much of the detail of a language's functional architecture is transmitted socially.

40.5.2 Co-Evolutionary Scenarios

In contemporary behavioral biology, the concept of instinct no longer comes with the connotation of learning playing no role. Many species-typical behaviors from the social learning of birdsongs to the hunting behavior of wolf packs involve the interaction of behavioral and learning biases with socially transmitted habits and variable environmental contexts. Darwin recognized the relevance of this environmental conditionality when he described the language adaptation as "an instinct to acquire an art." Language is, of course, special with regards to the relatively massive contribution of extrinsic factors, and also with respect to the likely combinatorial and emergent character of its supporting neurology. So its emergent character is unusually dependent on interactions between diverse neural and social mechanisms producing specific outcomes. This combinatorial co-dependence provides a challenge to simple caricatures of language evolution on the analogy of other physiological adaptations.

Recognition of this co-dependency has given rise to evolutionary scenarios that incorporate this interactional logic. Most develop from an evolutionary logic that has come to be called the Baldwin effect, after Baldwin (1896) who described how behavioral plasticity enabling the production of acquired adaptations might serve as an evolutionary precursor to a more innately produced analogue of this adaptation. The general logic of this evolutionary mechanism involves two phases: (1) the production of phenotypic plasticity (e.g., learned behaviors) making it possible for acquired adaptations to be conditionally produced that enable a lineage to persist despite a suboptimal match to the environment and (2) the appearance of new variants in that lineage that are selectively retained because they take over some fraction of the load of acquisition. This, presumably, described a Darwinian mechanism that would produce the evolutionary equivalent of Lamarckian inheritance of previously acquired traits. Proponents of innate UG invoked versions of this logic to argue that language-like behavior in our ancestors could have become progressively internalized as an innate faculty that is presently only minimally dependent on learning in the standard sense (e.g., Pinker, 1994). But the same logic could equally support the evolution of biases and aids to learning, without invoking a replacement of learned with innate knowledge of language (e.g., Deacon, 1992a, 1997). More recently, these arguments have been revisited in the context of the concept of niche construction (Odling-Smee *et al.*, 2003; Deacon, 2003), in which persistent socially maintained language use can be understood as a human-constructed niche that exerts significant selection pressures on the organism to adapt to its functional requirements. This approach is compatible with the claim that language function is supported by many modest distributed evolutionary modifications of brain anatomy and chemistry. It also assumes that language-like communication was present in some form for an extensive period of human prehistory.

40.5.3 Degenerative Processes as Possible Contributors to Language Evolution

A co-evolutionary scenario for the evolution of language still does not account for the generation of the novel functional synergy between neural systems that language processing requires. The discontinuities between call control systems and speech and language control systems of the brain suggest that a co-evolutionary logic alone is insufficient to explain the shift in substrate. Recent investigation of a

parallel shift in both complexity and neural substrate in birdsong may be able to shed some light on this.

In a comparative study of a long-domesticated bird, the Bengalese finch, and its feral cousin, the white-rump munia, it was discovered that the domesticated lineage was a far more facile song learner with a much more complex and flexible song than its wild cousin. This was despite the fact that the Bengalese Finch was bred in captivity for coloration, not singing (Okanoya, 2004). The domestic/feral difference of song complexity and song learning in these close finch breeds parallels what is found on comparisons between species that are song learners and nonlearners. This difference also correlates with a much more extensive neural control of song in birds that learn a complex and variable song. The fact that this behavioral and neural complexity can arise spontaneously without specific breeding for singing is a surprising finding since it is generally assumed that song complexity evolves under the influence of intense sexual selection. This was, however, blocked by domestication. One intriguing interpretation is that the relaxation of natural and sexual selection on singing paradoxically was responsible for its elaboration in this example. In brief, with the song becoming irrelevant to territorial defense, mate attraction, predator avoidance, and so on, degrading mutations and existing deleterious alleles affecting the specification of the stereotypic song would not have been weeded out. The result appears to have been the reduction of innate biases controlling song production. The domestic song could thus be described as both less constrained and more variable because it is subject to more kinds of perturbations. But with the specification of song structure no longer strictly controlled by the primary forebrain motor center (RA) (see Section 40.3.2), other linked brain systems can begin to play a biasing role. With the innate motor biases weakened, auditory experience, social context, learning biases, and attentional factors could all begin to influence singing. The result is that the domestic song became more variable, more complicated, and more influenced by social experience. The usual consequence of relaxed selection is genetic drift, increasing the genetic and phenotypic variety of a population by allowing random reassortment of alleles, but neurologically, drift in the genetic control of neural functions should cause constraints to become less specific, generating increased behavioral flexibility and greater conditional sensitivity to other neurological and contextual factors.

This is relevant to the human case, because a number of features of the human language adaptation also appear to involve a relaxation of innate constraints allowing multiple other influences besides fixed links to emotion and immediate context to affect vocalization. Probably the clearest evidence for this is infant babbling. This unprecedented tendency to freely play with vocal sound production occurs with minimal innate constraint on what sound can follow what (except for physical constraints on vocal sound generation). Babbling occurs also in contexts of comparatively low arousal state, whereas laughter, crying, or shrieking are each produced in comparatively specific high arousal states and with specific contextual associations. This reduction of innate arousal and contextual constraint on sound production opens the door for numerous other influences to begin to play a role. Like the domesticated bird, this allows many more brain systems to influence vocal behavior, including socially acquired auditory experience. In fact, this freedom from constraint is an essential precondition for being able to correlate learned vocal behaviors with the wide diversity of objects, events, properties, and relationships language is capable of referring to. It is also a plausible answer to the combinatorial synergy problem (discussed above) because it demonstrates an evolutionary mechanism that would spontaneously result in the emergence of multisystem coordination of neural control over vocal behavior.

But although an evolutionary dedifferentiation process may be a part of the story for human language adaptation, it is clearly not the whole story. This increased flexibility and conditionality likely exposed many previously irrelevant interrelationships between brain systems to selection for the new functional associations that have emerged. Most of these adaptations remain to be identified. However, if such a dedifferentiation effect has been involved in our evolution, then scenarios hypothesizing selection for increased innateness or extrapolation from innate referential calls to words become less plausible.

40.6 Conclusions

Despite decades of research to identify the distinctive neuroanatomical substrates that provide humans with an unprecedented faculty for language, no definitive core of uniquely human anatomical correlates has been demonstrated. Only a few distinctive anatomical differences can be directly associated with the human language adaptation. These are associated with the special motor adaptations for

speech. There is an unprecedented direct projection from motor cortex to the laryngeal motor nucleus of the brainstem (nucleus ambiguus) allowing direct control of vocalization independent of arousal state or innate vocal motor pattern. There is also one known genetic correlate with language competence, the gene FOXP2. Although it is clearly not specifically a language gene, nor can we be sure that its few human sequence differences represent adaptive modifications with respect to language, it is clear that it plays a necessary supportive role in the development of brain systems involved in speech production. Damage to gene in humans results in both generalized vocal disarthria and disruption of the ability to automate certain highly regular syntactic operations, and is associated with reduction of anterior basal ganglia structures. Besides these specific effects, however, it also appears likely that the neural changes associated with language adaptations involve more generalized allometric deviations from the ape pattern. Correlated with the increase in brain size in hominid evolution, there appears to have been quantitative remodeling of relationships between brain structures that is likely to have produced quantitative connectivity changes as well. If, as now appears likely, human brain adaptations for language involve many systems' coordinated interactions, it is likely that some or all of the quantitative alterations of brain organization reflect language adaptations. Although there is still considerable controversy concerning the proper assessment of the allometry human brain structures, candidates include overall cortical expansion, disproportion between cerebral cortex and basal ganglia, disproportionate increase in eulaminate cortical areas with respect to specialized sensory and motor areas, prefrontal expansion, increases in proportions of corticocortical and corticocerebellar connections, among others. However, the relevance of any of these cannot be discerned until there is a better understanding of the contributions of these systems to language acquisition, comprehension, and production. But a definitive assessment of the significant allometric deviations of human brain structure from typical primate trends could likewise provide hints of major differences in cognitive processing relevant to language.

The highly robust and developmentally canalized nature of language acquisition suggests that this capacity does not depend on only a few subtle neurological changes from the ape pattern but instead likely reflects a prolonged process of selection involving many systems, and perhaps extending over a million years. The nature of this selection process appears to have involved early protolanguage use as

a kind of niche construction, providing selective pressure to better support the unusual demands imposed by language. If this is an accurate assessment, it means that the neurological adaptations supporting language can at least in part be understood as adaptations for language, rather than merely accidentally giving rise to language. Some aspects of this ability may also be the result of evolutionary degradation of other functional specializations, which has allowed more diverse and distributed neural systems to directly or indirectly influence vocalization.

Though human brains unquestionably include numerous species-unique innate adaptations supporting the acquisition and use of language, there is to date little evidence for a specific neuroanatomical substrate for an UG. So, progress in understanding the language-related evolutionary changes of human brain structure can mostly be marked by what we now know is not the case, and just a few clear correlates of language adaptation. But this imposes considerable constraint on the scenarios we can consistently entertain and focuses neural research on a few notable problem areas.

References

- Aboitiz, F. and Garcia, V. 1997. The evolutionary origin of the language areas in the human brain: A neuroanatomical perspective. *Brain Res. Brain Res. Rev.* 25, 381–396.
- Amunts, K., Schleicher, A., Burgel, U., Mohlberg, H., Uylings, H. B. M., and Zilles, K. 1999. Broca's region revisited: Cytoarchitecture and inter subject variability. *J. Comp. Neurol.* 412, 319–341.
- Baldwin, J. M. 1896. A new factor in evolution. *Am. Nat.* 30, 441–451, 536–553.
- Blakeslee, P. and Gunter, R. 1966. Cross modal transfer of discrimination learning in Cebus monkeys. *Behaviour* 26(1), 76–90.
- Boatman, D., Freeman, J., Vining, E., et al. 1999. Language recovery after left hemispherectomy in children with late onset seizures. *Ann. Neurol.* 46(4), 579–586.
- Broca, P. 1865. Sur la faculté du langage articulé. *Bull. Soc. Anthropol. (Paris)* 6, 337–393.
- Brodmann, K. 1909. Vergleichende Lokalisationlehre der Grosshirnrinde. Barth.
- Catani, M., Jones, D. K., and ffytche, D. H. 2005. Perisylvian language networks of the human brain. *Ann. Neurol.* 57(1), 8–16.
- Chomsky, N. 1972. Language and Mind. Harcourt Brace Jovanovich.
- Corballis, M. 2002. From Hand to Mouth: The Origins of Language. Princeton University Press.
- Darwin, C. 1871. The Descent of Man, and Selection in Relation to Sex. John Murray.
- Deacon, T. W. 1988. Human brain evolution. II: Embryology and brain allometry. In: Intelligence and Evolutionary Biology (eds. H. J. Jerison and I. Jerison), pp. 383–416. Springer.
- Deacon, T. 1990a. Fallacies of progression in theories of brain size evolution. *Int. J. Primatol.* 11, 193–236.
- Deacon, T. 1990b. Rethinking mammalian brain evolution. *Am. Zool.* 30, 629–705.
- Deacon, T. 1992a. Brain language coevolution. In: The Evolution of Human Languages (eds. A. John, J. A. Hawkins, and M. Gel Mann), pp. 49–83. Addison Wesley.
- Deacon, T. 1992b. Cortical connections of the inferior arcuate sulcus cortex in the macaque brain. *Brain Res.* 573, 8–26.
- Deacon, T. 1997. The Symbolic Species: The Coevolution of Language and the Brain. W. W. Norton.
- Deacon, T. 2003. Multilevel selection in a complex adaptive system: The problem of language origins. In: Evolution and Learning: The Baldwin Effect Reconsidered (eds. B. Weber and D. Depew), pp. 81–106. MIT Press.
- Deacon, T. 2004. Monkey homologues of language areas: Computing the ambiguities. *Trends Cogn. Sci.* 8, 288–289.
- D'Esposito, M. and Alexander, M. P. 1995. Subcortical aphasia: Distinct profiles following left putaminal hemorrhage. *Neurology* 45, 38–41.
- DeVoogd, T., Krebs, J. R., Healy, S. D., and Purvis, A. 1993. Relations between song repertoire size and the volume of brain nuclei related to song: Comparative evolutionary analyses among oscine birds. *Proc. R. Soc. Lond. B* 254, 75–82.
- Dronkers, N. F., Shapiro, J. K., Redfern, B., and Knight, R. T. 1992. The role of Broca's area in Broca's aphasia. *J. Clin. Exp. Neuropsychol.* 14, 198.
- Dunbar, R. I. M. 1993. Coevolution of neocortical size and language in humans. *Behav. Brain Sci.* 16, 681–735.
- Enard, W., Przeworski, M., Fisher, S. E., et al. 2002. Molecular evolution of FOXP2, a gene involved in speech and language. *Nature* 418, 869–872.
- Ettlinger, G. and Wilson, W. A. 1990. Cross modal performance: Behavioural processes, phylogenetic considerations, and neural mechanisms. *Behav. Brain Res.* 40(3), 169–192.
- Falk, D. 1983. Cerebral cortices of East African early hominids. *Science* 221, 1072–1074.
- Finlay, B. L. and Darlington, R. B. 1995. Linked regularities in the development and evolution of mammalian brains. *Science* 268, 1578–1584.
- Gannon, P. J., Holloway, R. L., Broadfield, D. C., and Braun, A. R. 1998. Asymmetry of chimpanzee planum temporale: Humanlike pattern of Wernicke's brain language area homolog. *Science* 279, 220–222.
- Gardner, H., Brownell, H. H., Wapner, W., and Michelow, D. 1983. Missing the point: The role of the right hemisphere in processing of complex linguistic materials. In: Cognitive Processes and the Right Hemisphere (ed. E. Perecman). Academic Press.
- Geschwind, N. 1964. The development of the brain and the evolution of language. *Monogr. Ser. Lang. Linguist.* 1, 155–169.
- Gopnik, M. 1990. Feature blind grammar and dysphasia in an extended family. *Nature* 344, 715.
- Gopnik, M. and Crago, M. 1991. Familial segregation of a developmental language disorder. *Cognition* 39, 1–50.
- Gould, S. J. 1975. Allometry in primates, with emphasis on scaling and the evolution of the brain. In: Approaches to Primate Paleobiology (ed. Szalay). Karger.
- Haesler, S., Wada, K., Nshdejan, A., et al. 2004. FoxP2 expression in avian vocal learners and non learners. *J. Neurosci.* 24(13), 3164–3175.
- Hewes, G. W. 1973. Primate communication and the gestural origin of language. *Curr. Anthropol.* 14, 5–24.

- Holloway, R. 1979. Brain size, allometry, and reorganization: Toward a synthesis. In: *Development and Evolution of Brain Size* (eds. M. Hahn, C. Jensen, and B. Dudek). Academic Press.
- Jackendoff, R. 1994. *Patterns in the Mind: Language and Human Nature*. Basic Books.
- Jarvis, E. 2004. Learned birdsong and the neurobiology of human language. *Ann. NY Acad. Sci.* 1016, 749–777.
- Johansson, S. 2005. *Origins of Language: Constraints on Hypotheses*. John Benjamins.
- Jurgens, U. 1979. Neural control of vocalization in non human primates. In: *Neurobiology of Social Communication in Primates* (eds. H. D. Steklis and M. J. Raleigh). Academic Press.
- Jurgens, U., Kirzinger, A., and von Cramon, D. 1982. The effects of deep reaching lesions in the cortical face area on phonation. A combined case report and experimental monkey study. *Cortex* 18, 125–139.
- Kimura, D. 1993. *Neuromotor Mechanisms in Human Communication*. Oxford University Press.
- Kolb, B. 1995. *Brain Plasticity and Behavior*. Lawrence Erlbaum.
- Lai, C. S., Gerrelli, D., Monaco, A. P., Fisher, S. E., and Copp, A. J. 2003. FOXP2 expression during brain development coincides with adult sites of pathology in a severe speech and language disorder. *Brain* 126, 2455–2462.
- Lenneberg, E. H. 1967. *Biological Foundations of Language*. Wiley.
- Lieberman, P. 2002. On the nature and evolution of the neural bases of human language. *Yrbk. Phys. Anthropol.* 45, 36–62.
- Neville, H., Coffey, S., Lawson, D., Fischer, A., Emmorey, K., and Bellugi, U. 1997. Neural systems mediating American Sign Language. *Brain Lang.* 57, 285–308.
- Nottebohm, F. and Nottebohm, M. E. 1976. Left hypoglossal dominance in the control of canary and white crowned sparrow song. *J. Comp. Physiol.* 108, 171–192.
- Odling Smee, F. J., Laland, K. N., and Feldman, M. W. 2003. *Niche Construction: The Neglected Process in Evolution*. Princeton University Press.
- Okanoya, K. 2004. The bengalese finch: A window on the behavioral neurobiology of birdsong syntax. *Ann. NY Acad. Sci.* 1016, 724–735.
- Paulesu, E., et al. 1997. Functional heterogeneity of left inferior frontal cortex as revealed by fMRI. *Neuroreport* 8, 2011–2016.
- Pinker, S. 1994. *The Language Instinct: How the Mind Creates Language*. William Morrow.
- Provine, R. 2000. *Laughter: A Scientific Investigation*. Viking.
- Rilling, J. and Seligman, R. 2002. A quantitative morphometric comparative analysis of the primate temporal lobe. *J. Hum. Evol.* 42, 505–533.
- Rilling, J. K. and Insel, T. R. 1999. The primate neocortex in comparative perspective using magnetic resonance imaging. *J. Hum. Evol.* 37, 191–223.
- Rizzolatti, G. and Arbib, M. A. 1998. Language within our grasp. *Trends Neurosci.* 21, 188–194.
- Romanski, L., et al. 1999. Dual streams of auditory afferents target multiple domains in the primate prefrontal cortex. *Nat. Neurosci.* 2, 1131–1136.
- Ross, E. D. 1981. The aprosodias: Functional anatomical organization of the affective components of language in the right hemisphere. *Arch. Neurol.* 36, 561–569.
- Sacher, G. A. 1970. Allometric and factorial analysis of brain structure in insectivores and primates. In: *The Primate Brain* (eds. C. R. Noback and W. Montagna). Appleton Century Crofts.
- Semendeferi, K., Damasio, H., Frank, R., and Van Hoesen, G. W. 1997. The evolution of the frontal lobes: A volumetric analysis based on three dimensional reconstructions of magnetic resonance scans of human and ape brains. *J. Hum. Evol.* 32, 375–378.
- Seyfarth, R., Dorothy, C., and Peter, M. 1980. Monkey responses to three different alarm calls: Evidence of predator classification and semantic communication. *Science* 210, 801–803.
- Shua, W., Chob, J., Jiang, Y., et al. 2005. Altered ultrasonic vocalization in mice with a disruption in the Foxp2 gene. *Proc. Natl. Acad. Sci.* 102, 9643–9648.
- Stephan, H. 1969. Quantitative investigations on visual structures in primate brains. In: *Proceedings of the Second International Congress on Primates*, No. 3, pp. 34–42. Karger.
- Stephan, H. H., Frahm, B., and Baron, G. 1981. New and revised data on volumes of brain structures in insectivores and primates. *Folia Primatol. (Basel)* 35, 1–29.
- Tobias, P. V. 1987. The brain of *Homo habilis*: A new level of organisation in cerebral evolution. *J. Hum. Evol.* 16, 741–761.
- Vargha Khadem, F., Watkins, K. E., Price, C. J., et al. 1998. Neural basis of an inherited speech and language disorder. *Proc. Natl. Acad. Sci.* 95, 12695–12700.
- Wegener, J. G. 1965. Cross modal transfer in monkeys. *J. Comp. Physiol. Psychol.* 59, 450–452.
- Wernicke, C. 1874. *The Aphasic Symptom Complex: A Psychological Study on a Neurological Basis*. Kohn and Weigert (eds. R. S. Cohen and M. W. Wartofsky). Reprint, Boston Studies in the Philosophy of Science, vol. 4. Reidel.

Further Reading

- Christiansen, M. and Kirby, S. (eds.) 2004. *Language Evolution*. Oxford University Press.